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KIRRIBILLI 2061



**KINCOPPAL-ROSE BAY
SCHOOL OF THE SACRED HEART**

TRIAL EXAMINATION

YEAR 12

2000

MATHEMATICS

3 UNIT (ADDITIONAL)

AND

3/4 UNIT (COMMON)

Time allowed - Two hours
(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on the last page
- Board approved calculators may be used
- Answer each question in a SEPARATE Writing Booklet
- You may ask for extra Writing Booklets if you need them.

QUESTION 1 (Start a new Booklet)

Marks

(a) Solve $\frac{x^2 - 4}{x} \geq 3$

3

(b) (i) Sketch the graph of $y = |2 - 3x|$

(ii) Hence, or otherwise, solve $|2 - 3x| < x$

3

(c) Find the acute angle between the lines
 $3x - 2y - 5 = 0$ and $x - 5y - 3 = 0$

2

(d) On the number plane, A and B are points with co-ordinates $(5, 3)$ and $(1, -3)$ respectively. P is the point on AB which divides the interval AB externally in the ratio $3:2$.

(i) Show that the co-ordinates of P are $(-7, -15)$.

(ii) C is the point such that the line through P which is parallel to BC meets the line AC at $Q(8, -12)$. Find the co-ordinates of C .

4

QUESTION 2

(Start a new Booklet)

Marks

(a) Evaluate the following:

8

$$(i) \int_0^2 \frac{dx}{1 + (x-1)^2}$$

$$(ii) \int \frac{e^x dx}{\sqrt{1 - e^{2x}}}$$

$$(iii) \int_0^{\pi/2} \frac{\cos x dx}{1 + \sin^2 x} \text{ using } u = \sin x$$

(b) Show that $\cos(A+B) + \cos(A-B) = 2 \cos A \cos B$ Hence or otherwise, evaluate $\int_0^{\pi/6} 2 \cos 3x \cos x dx$

4

QUESTION 3**(Start a new Booklet)****Marks**

- (a) (i) Assuming $\cos x \neq 0$, make $\tan x$ the subject of $\sin(x + \theta) = a \cos x$.
- (ii) Use the result from (i) to find the exact value of $\tan x$ when $\sin(x + \frac{\pi}{3}) = 2 \cos x$ and the value(s) of x , $0 \leq x \leq 2\pi$ correct to 4 decimal places.
- (b) Differentiate $\ln \left\{ \frac{1 - \sin x}{1 + \sin x} \right\}$ with respect to x .
Express your answer in simplest form.
- (c) Show that $\lim_{h \rightarrow 0} \frac{1 - \cos 2h}{h^2} = 2$
- (d) Prove by Mathematical Induction that $2^n - 1$ is divisible by 7.

4

3

2

3

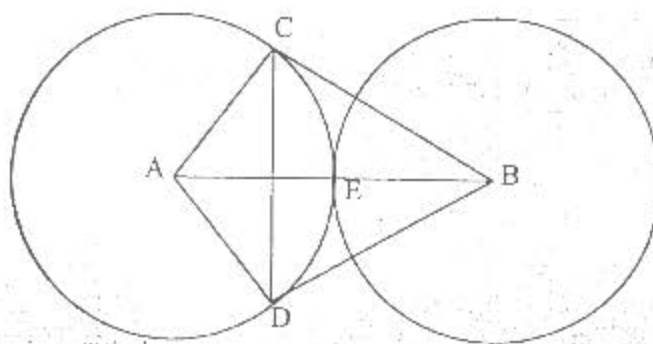
QUESTION 4

(Start a new Booklet)

Marks

(a)

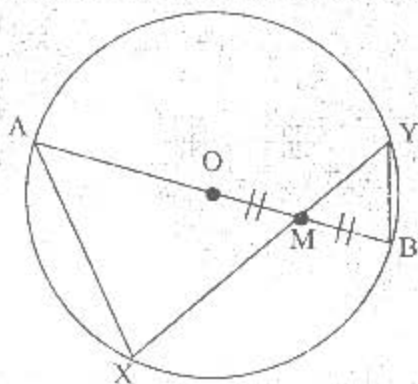
8



Two circles of equal radius and with centres at A and B respectively touch each other externally at E. BC and BD are tangents from B to the circle with centre A.

- Copy the diagram.
- Show that BCAD is a cyclic quadrilateral.
- Show that E is the centre of the circle which passes through B, C, A and D.
- Show that $\angle CBA = \angle DBA = 30^\circ$.
- Show that triangle BCD is equilateral.

- (b) In the diagram below, AB is a diameter of a circle, whose centre is the point O. The chord XY passes through M, the mid point of OB. AX and BY are joined.



- Prove the two triangles formed (triangles AXM and MYB) are similar. If $XM = 8$ cm and $YM = 6$ cm, find the length of the radius of the circle.

QUESTION 5

(Start a new Booklet)

Marks

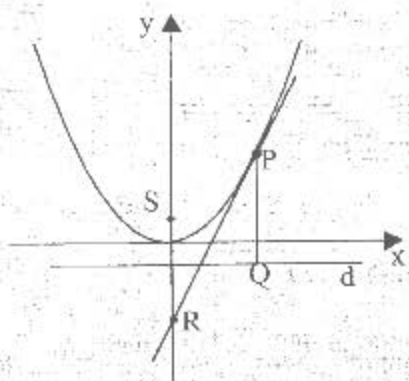
- (a) (i) Show that the normal to the parabola $x^2 = 4ay$ at the point $T(2at, at^2)$ has equation

$$x + ty = 2at + at^3.$$

- (ii) Hence show that there is only one normal to the parabola which passes through its focus.

4

(b)



$P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$. S is the focus of the parabola. PQ is the perpendicular from P to the directrix d of the parabola. The tangent at P to the parabola cuts the axis of the parabola at the point R .

- (i) Show that the tangent at P to the parabola has equation

$$tx - y - at^2 = 0.$$

- (ii) Show that PR and QS bisect each other.

- (iii) Show that PR and QS are perpendicular to each other. State with reason what type of quadrilateral $PQRS$ is.

8

QUESTION 6 (Start a new Booklet)

Marks

- (a) You are given that 0.8 is an approximate root of the equation $e^{-x} - 0.5x = 0$. Using one application of Newton's Method, find a better approximation, correct to 3 decimal places.

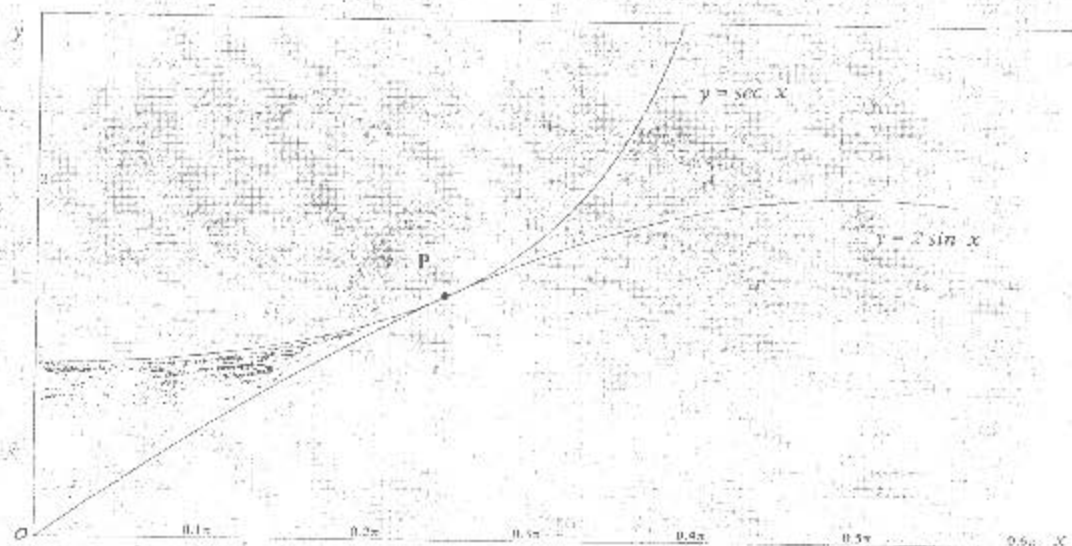
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- (b) (i) Differentiate $\cos^4 x$.

3

- (ii) Hence, find the exact value of $\int_0^{\frac{\pi}{4}} \sin x \cos^3 x \, dx$.

(c)



P is the point of intersection of the graphs $y = \sec x$ and $y = 2 \sin x$ in the domain $0 < x < \frac{\pi}{2}$.

6

- (i) Verify that P is the point $\left(\frac{\pi}{4}, \sqrt{2}\right)$.
- (ii) The shaded region is rotated about the x -axis. Find the volume of the solid of revolution formed.

QUESTION 7**(Start a new Booklet)****Marks**

- (a) If α, β, γ are the roots of $5x^3 - 2x^2 - 4x + 7 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$. 2

- (b) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos (x + \alpha)$, $R > 0$. 4

- (ii) Hence, find the general solutions for $\sqrt{3} \cos x - \sin x = 1$.

- (c) (i) Sketch the curve $y = \cos^{-1} 2x$. 6

- (ii) State the domain and range of the function.

- (iii) Find the equation of the tangent to the curve $y = \cos^{-1} 2x$ at the point where the curve cuts the y -axis.

The End

Question 1

MINICORPAL
30 TRIAL SOLNS

2000

(a)

$$\frac{x^2 - 4}{x} > 3$$

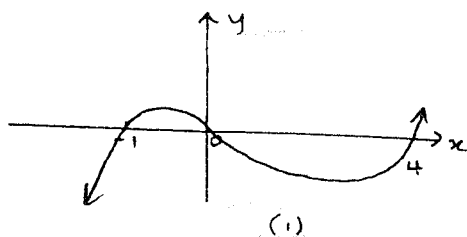
⊗ x^2

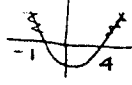
$$x(x^2 - 4) > 3x^2$$

$$x(x^2 - 3x - 4) > 0$$

$$x(x - 4)(x + 1) > 0 \quad (1)$$

(3)



$x \neq 0$	
$x > 0$	$x < 0$
$x^2 - 4 > 3x$	$x^2 - 4 \leq 3x$
$(x^2 - 3x - 4) > 0$	
$(x - 4)(x + 1) > 0$	
	$-1 \leq x \leq 4$
$x \leq -1 \text{ or } x > 4$	
$x > 4$	$-1 \leq x < 0$

Solution:

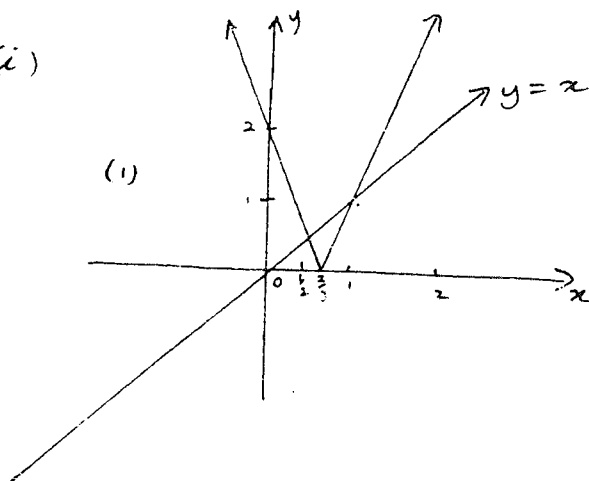
(1) $x > 4$,

$-1 \leq x < 4$

(3)

(b)

(i)



$$2 - 3x = 0$$

$$2 = 3x$$

$$x = \frac{2}{3}$$

Critical pts:

$$2 - 3x = x$$

$$2 = 4x$$

$$\frac{2}{4} = x$$

$$\therefore x = \frac{1}{2}$$

$$-(2 - 3x) = x$$

$$-2 + 3x = x$$

$$-2 = -2x$$

$$x = 1$$

(ii) $|2 - 3x| < x$

(2)

from graph

$$\frac{1}{2} < x < 1$$

(2)

$$(c) \quad 3x - 2y - 5 = 0$$

$$3x - 5 = 2y$$

$$y = \frac{3}{2}x - \frac{5}{2}$$

$$\therefore m_1 = \frac{3}{2}$$

$$x - 5y - 3 = 0$$

$$x - 3 = 5y$$

$$y = \frac{1}{5}x - \frac{3}{5}$$

$$m_2 = \frac{1}{5}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

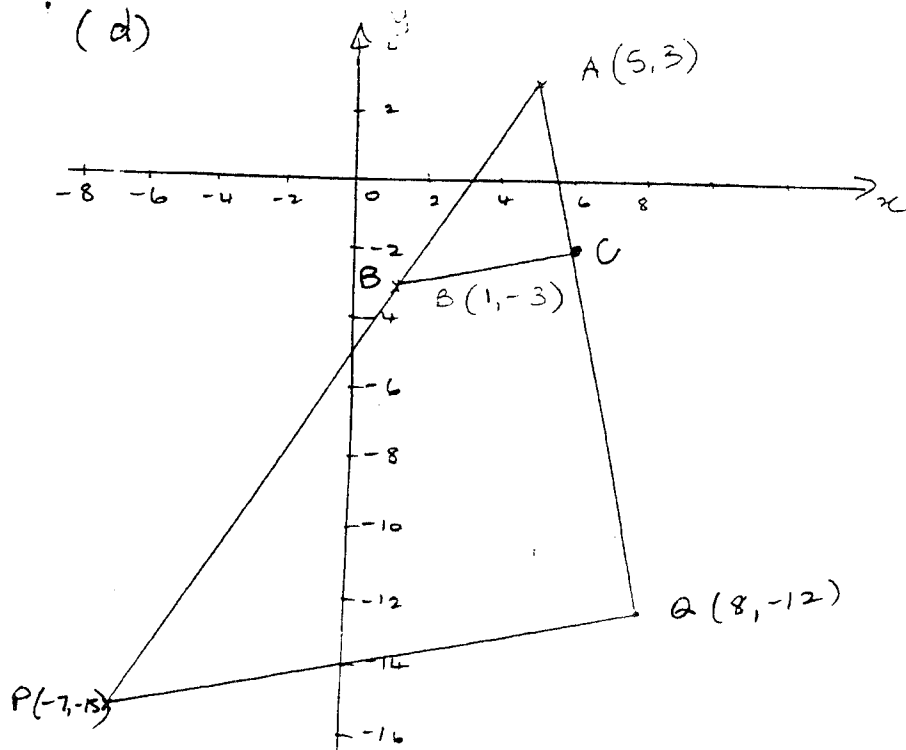
$$= \left| \frac{\frac{3}{2} - \frac{1}{5}}{1 + \frac{3}{2} \cdot \frac{1}{5}} \right|$$

$$= 1$$

$$\therefore \underline{\theta = 45^\circ}$$

(2) marks

(d)



i)
$$\begin{array}{ccc} -3 & \times & 5 \\ 2 & \times & 1 \end{array}, \begin{array}{c} 3 \\ -3 \end{array}$$

$$\left(\frac{-3 \times 1 + 2 \times 5}{-3 + 2}, \frac{-3 \times -3 + 2 \times 3}{-3 + 2} \right)$$

$$= \left(\frac{-3 + 10}{-1}, \frac{9 + 6}{-1} \right)$$

$$= (-7, -15)$$

ii) Since P divides AB externally in ratio of 3:2, then B divides AP internally in ratio 1:2.

Since PQ $\parallel$ BC, C divides AQ internally in ratio 1:2. A(5, 3), Q(8, -12)

$$\begin{array}{ccc} 1 & \times & 5 \\ 2 & \times & 8 \end{array}, \begin{array}{c} 3 \\ -12 \end{array} \quad \left(\frac{1 \times 8 + 2 \times 5}{3}, \frac{1 \times -12 + 2 \times 3}{3} \right)$$
$$= \left(\frac{18}{3}, \frac{-12 + 6}{3} \right) = (6, -2)$$

QUESTION 2

$$\begin{aligned}
 (a) (i) \int_0^2 \frac{dx}{1+(x-1)^2} &= \left[\tan^{-1}(x-1) \right]_0^2 \\
 &= \tan^{-1} 1 - \tan^{-1}(-1) \\
 &= -\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$(ii) \int \frac{e^x dx}{\sqrt{1-e^{2x}}} = \sin^{-1}(e^x) + C$$

$$(iii) \int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x} dx$$

$$= \int_0^1 \frac{du}{1+u^2}$$

$$= \left[\tan^{-1} u \right]_0^1$$

$$= \tan^{-1} 1 - \tan^{-1} 0$$

$$= \frac{\pi}{4} - 0$$

$$= \frac{\pi}{4}$$

using

$$u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$du = \cos x dx$$

$$\text{when } x = \frac{\pi}{2}, u =$$

$$x=0, u=0$$

2 (b)

$$\begin{aligned} \text{LHS} &= \cos A \cos B - \sin A \sin B + \cos A \cos B + \sin A \sin B \\ &= 2 \cos A \cos B \end{aligned}$$

$$\therefore \int_0^{\pi/6} 2 \cos 3x \cos x \, dx$$

$$= \int_0^{\pi/6} \cos 4x + \cos 2x \, dx$$

$$= \left[\frac{1}{4} \sin 4x + \frac{1}{2} \sin 2x \right]_0^{\pi/6}$$

$$= \frac{3\sqrt{3}}{8}$$

$$(a)(i) \sin(x+\theta) = a \cos x$$

$$\sin x \cos \theta + \sin \theta \cos x = a \cos x$$

$$\div \cos x, \cos x \neq 0$$

$$\tan x \cos \theta + \sin \theta = a$$

$$\tan x = \frac{a - \sin \theta}{\cos \theta} \quad (2)$$

(ii)

$$\tan x = \frac{2 - \sin \frac{\pi}{3}}{\cos \frac{\pi}{3}}$$

$$= \frac{2 - \frac{\sqrt{3}}{2}}{\frac{1}{2}} \quad (1)$$

$$= 4 - \sqrt{3}$$

$$\text{Hence } x = \tan^{-1}(4 - \sqrt{3}), \quad 0 \leq x \leq 2\pi$$

$$= 1.1555, \quad 4.2971 \quad (2)$$

$$(b) \quad y = \ln \left(\frac{1 - \sin x}{1 + \sin x} \right) = \ln(1 - \sin x) - \ln(1 + \sin x)$$

$$\text{let } f(x) = \frac{1 - \sin x}{1 + \sin x}$$

$$= \frac{-\cos x}{1 - \sin x} - \frac{\cos x}{1 + \sin x}$$

$$= \frac{-\cos x - \sin x \cos x - \cos x + \sin x \cos x}{(1 - \sin x)(1 + \sin x)}$$

$$\therefore f'(x) = \frac{(1 + \sin x)(-\cos x) - (1 - \sin x)(\cos x)}{(1 + \sin x)^2}$$

$$= \frac{-2 \cos x}{1 + \sin x} = -2 \sec x$$

$$f'(x) = \frac{-\cos x - \cos x}{(1 + \sin x)^2}$$

$$= \frac{-2\cos x}{(1 + \sin x)^2}$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{-2\cos x}{(1 + \sin x)^2} \times \frac{1 + \sin x}{1 - \sin x}$$

$$= \frac{-2\cos x}{1 - \sin^2 x}$$

$$= -2\sec x$$

$$(c) \lim_{h \rightarrow 0} \frac{1 - (1 - 2\sin^2 h)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{2\sin^2 h}{h^2}$$

$$= 2 \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \frac{\sin h}{h}$$

$$= 2 \times 1 \times 1$$

$$= 2$$

(d) Step 1: When $n=1$, $2^{3n}-1 = 2^3-1 = 7$
which is divisible by 7.

(1) Thus the statement is true for $n=1$

Step 2: Assume true for $n=k$

ii: $2^{3k}-1 = 7m$ where m is
any integer

Step 3: { Prove true for $n=k+1$

(1) ie: $2^{3k+3}-1$ is divisible by 7.

Now $2^{3k+3}-1 = 2^{3k} \cdot 2^3 - 1$

$$= (7m+1) 2^3 - 1$$

Since $2^{3k} = 7m+1$ from Step:

$$= (7m+1) 8 - 1$$

$$= 56m + 8 - 1$$

$$= 56m + 7$$

$$(1) = 7(8m+1)$$

(2) $= 7p$ which is divisible by
where $p=8m+1$ and $8m+1$
is an integer.

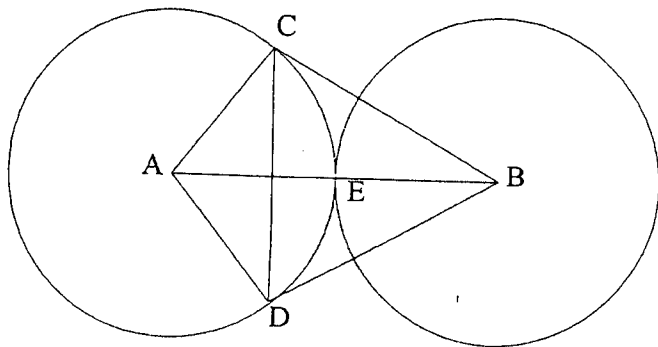
Thus $2^{3(k+1)}-1$ is divisible by 7.

Thus if it is true for $n=k$, it is true for
 $n=k+1$. It is true for $n=1$, then it is true
for $n=2$, and so it is true for all positive n .

Solutions and Marking Scheme

Question 4

(a) i.



ii.

 $\angle BCA = \angle BDA = 90^\circ$ (BC and BD are tangents to circle with centre A)

$$\therefore \angle BCA + \angle BDA = 180^\circ$$

 $\angle BCA + \angle BDA + \angle CAD + \angle CBD = 360^\circ$ (Angle sum of a quadrilateral)

$$\therefore 180^\circ + \angle CAD + \angle CBD = 360^\circ$$

$$\therefore \angle CAD + \angle CBD = 360^\circ - 180^\circ = 180^\circ$$

 $\therefore$ BCAD is a cyclical quadrilateral (Opposite angles add to 180°) (2 marks)

(iii)

$$\angle BCA = \angle BDA = 90^\circ \text{ (Above)}$$

 $\therefore$ AB is the diameter of the circle which passes through B, C, A and D
(angle in a semicircle is 90°)
Now $AE = BE$

(circle A and circle B have the equal radii, given)

 $\therefore$ E is the MP of AB (the diameter)

 $\therefore$ E is the centre of the circle.

(2 marks)

(iv)

$$AE = BE \text{ (above)}$$

$$AC = AE \text{ (equal radii)}$$

$$AB = 2AC$$

$$\angle BCA = 90^\circ \text{ (above)}$$

In $\triangle ABC$,

$$\sin(\angle CBA) = \text{Opp} / \text{Adj}$$

$$= AC / AB$$

$$= AC / 2AC$$

$$= \frac{1}{2}$$

$$\therefore \angle CBA = 30^\circ$$

Similarly in $\triangle ABD$,

$$\angle DBA = 30^\circ.$$

$$\text{or } AC = AE \text{ (radii)}$$

$$AE = EC \text{ (radii circle centre E)}$$

 $\therefore \triangle ACE$ is equilateral

$$\therefore \angle CAE = 60^\circ$$

 $\therefore \angle ACB = 90^\circ$ from before

$$(2 \text{ marks}) \quad \angle CBA = 30^\circ$$

(angle sum $\triangle$)

$$\angle CBA = \angle ABD$$

since AB is line of symmetry

$$\begin{aligned} \text{v. } \angle CBD &= \angle CBA + \angle DBA \\ &= 30^\circ + 30^\circ \\ &= 60^\circ \end{aligned}$$

$$\angle BCD + \angle CDB + \angle CBD = 180^\circ \text{ (angle sum of a triangle)}$$

$$\angle BCD + \angle CDB + 60^\circ = 180^\circ$$

$$\therefore \angle BCD + \angle CDB = 120^\circ$$

But $\angle BCD = \angle CDB$ (base angles of isosceles triangle, $BC = BD$, tangents to a circle from an external point are equal in length)

$$\therefore \angle BCD = \angle CDB = 60^\circ$$

$$\therefore \triangle CBD \text{ is equilateral (all angles are equal)} \quad (2 \text{ marks})$$

$$\begin{aligned} \text{(b) i. } \angle BAX &= \angle XYB \text{ (Angles subtended by common arc XB)} \\ \angle AXY &= \angle YBA \text{ (Angles subtended by common arc AY)} \\ \angle AMX &= \angle BMY \text{ (Vertically opposite angles)} \\ \therefore \triangle AXM &\parallel \triangle MYB \text{ (AAA)} \quad (2 \text{ marks}) \end{aligned}$$

We know that

$$AB = 2OB \text{ (diameter = 2 x radius)}$$

$$OB = 2BM \text{ (M is MP of OB)}$$

$$\therefore AB = 4BM \text{ and } AM = 3BM$$

Corresponding sides of similar triangles are in a common ratio.

$$\therefore XM/BM = AM/YM$$

$$\text{ie } 8/BM = AM/6$$

$$\text{or } 6 \times 8 = BM \times AM$$

$$\text{or } 48 = BM \times (3BM)$$

$$\therefore 48/3 = BM \times BM$$

$$BM = 4 \text{ cm and the radius} = 2BM = 8 \text{ cm} \quad (2 \text{ marks})$$

$$\triangle AXM \parallel \triangle MYB$$

$$\frac{XM}{YB} = \frac{AX}{MY}$$

$$\frac{AM}{YM} = \frac{XM}{MB}$$

$$\frac{8}{x}$$

$$\frac{3x}{6} = \frac{8}{x}$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = 4$$

$$\text{radius} = 8 \text{ cm}$$

Question 5

(a) i. $x^2 = 4ay$
 $\therefore y = x^2 / 4a$
 $dy/dx = 2x/4a = x/2a$
 $\therefore m = x/2a = t$ (when $x = 2at$)
 $\therefore m \text{ of normal} = -1/t$
 $(x_1, y_1) = (2at, at^2)$

now $y - y_1 = (m \text{ of normal}) (x - x_1)$
or $y - at^2 = -1/t (x - 2at)$
 $ty - at^3 = -x + 2at$
 $\therefore x + ty = 2at + at^3$. (2 marks)

For the normal to pass through the focus, $(0, a)$ must satisfy the equation of the normal.
ie sub $(0, a)$ into $x + ty = 2at + at^3$

$$0 + ta = 2at + at^3$$

$$0 = at + at^3$$

$$at + at^3 = 0$$

$$at(1 + t^2) = 0$$

there is only one real solution: $t = 0$.

$\therefore$ there is only one normal to the parabola which passes through its focus. (2 marks)

(b) i. At $P(2at, at^2)$, $y = x^2/4a$, $\therefore dy/dx = 2x/4a = x/2a$ and $m = 2at/2a = t$
Equation of tangent at P:
 $y - y_p = m(x - x_p)$
 $y - at^2 = t(x - 2at)$
 $y - at^2 = tx - 2at^2$
 $\therefore tx - y - at^2 = 0$ (1) (2 marks)

ii **P(2at, at²)**

Directrix has equation $y = -a$ and thus **Q(2at, -a)**

R intersects the y-axis and so $x = 0$ at R and substituting this value into (1) yields:

$$-y - at^2 = 0$$

$$\therefore y = -at^2 \text{ and } \mathbf{R(0, -at^2)}$$

S is the focus ie **S(0,a)**.

$$\text{Gradient of PR} = m_{PR} = t \text{ (from above)}$$

$$\text{Equation of PR: } tx - y - at^2 = 0$$

$$\therefore y = tx - at^2 \quad (2)$$

$$\text{Gradient of QS} = m_{QS} = (y_Q - y_S) / (x_Q - x_S)$$

$$= (-a - a) / (2at - 0)$$

$$= -2a / 2at$$

$$= -1/t$$

$$\text{Equation of QS: } y - y_S = m_{QS} (x - x_S)$$

$$y - a = -1/t(x - 0)$$

$$\therefore y = -x/t + a \quad (3)$$

Let Intersection of QS and PR be $M(x_M, y_M)$

For M, set (2) = (3)

$$tx - at^2 = -x/t + a$$

$$t^2x - at^3 = -x + at$$

$$t^2x + x = at^3 + at$$

$$x(t^2 + 1) = at(t^2 + 1)$$

$$\therefore x_M = at \quad (4)$$

$$\text{Sub (4) into (3): } \therefore y_M = -at/t + a$$

$$= -a + a$$

$$= 0$$

$$\therefore M(x_M, y_M) = \mathbf{M(at, 0)}$$

$$\begin{aligned} \text{now the midpoint of QS} &= MP_{QS} \left(\frac{1}{2}(x_Q + x_S), \frac{1}{2}(y_Q + y_S) \right) \\ &= MP_{QS} \left(\frac{1}{2}(2at + 0), \frac{1}{2}(-a + a) \right) \\ &= MP_{QS} (at, 0) \\ &= M(at, 0) \end{aligned}$$

$$\begin{aligned} \text{Similarly the midpoint of PR} &= MP_{PR} \left(\frac{1}{2}(x_Q + x_S), \frac{1}{2}(y_Q + y_S) \right) \\ &= MP_{PR} \left(\frac{1}{2}(2at + 0), \frac{1}{2}(at^2 - at^2) \right) \\ &= MP_{PR} (at, 0) \\ &= M(at, 0) \end{aligned}$$

PQ and RS bisect each other at M since M is the midpoint of both PQ and RS.
(4 marks)

iii Gradient of PR = $m_{PR} = t$ (from above)

$$\text{Gradient of QS} = m_{QS} = -1/t \text{ (from above)}$$

$$\text{So } m_{QS} = -1/t$$

$$= -1/m_{PR}$$

$\therefore$ PR is perpendicular to QS. (1 mark)

iv. PQRS is a rhombus since the diagonals QS and PR bisect each other at right angles.
(1 mark)

Question 6 solutions

$$(a) \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{Now } y = e^{-x} - 0.5x$$

$$y' = -e^{-x} - 0.5$$

$$x_2 = 0.8 - \frac{(e^{-0.8} - 0.5(0.8))}{(-e^{-0.8} - 0.5)}$$

$$= 0.8 - \frac{(e^{-0.8} - 0.4)}{(-e^{-0.8} - 0.5)}$$

$$\frac{0.04932\dots}{-0.9493\dots}$$

$$= 0.8 - (-0.0519619\dots)$$

$$= 0.8519619\dots$$

$$\therefore x_2 = 0.852 \quad (3 \text{ D.P.})$$

Question 6 Solutions

$$(a) \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\text{Now } y = e^{-x} - 0.5x$$

$$y' = -e^{-x} - 0.5$$

$$x_2 = 0.8 - \frac{(e^{-0.8} - 0.5(0.8))}{(-e^{-0.8} - 0.5)}$$

$$= 0.8 - \frac{(e^{-0.8} - 0.4)}{(-e^{-0.8} - 0.5)}$$

$$\frac{0.04932...}{-0.9493...}$$

$$= 0.8 - (-0.0519619...)$$

$$= 0.8519619...$$

$$\therefore x_2 = 0.852 \quad (3 \text{ D.P.})$$

$$(c) \quad (i) \quad \text{When } x = \frac{\pi}{4}, \quad y = \sec \frac{\pi}{4} \\ = \sqrt{2}$$

$$\text{When } x = \frac{\pi}{4}, \quad y = 2 \sin \frac{\pi}{4} \\ = 2 \times \frac{1}{\sqrt{2}} \\ = \frac{2}{\sqrt{2}} \\ = \sqrt{2}$$

$\therefore P^1 \left(\frac{\pi}{4}, \sqrt{2} \right)$ is a pt of int.

$$(ii) \quad V = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx - \pi \int_0^{\frac{\pi}{4}} (2 \sin x)^2 \, dx \\ = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx - 4\pi \int_0^{\frac{\pi}{4}} \sin^2 x \, dx \\ = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx - 4\pi \left(\frac{1}{2} \int_0^{\frac{\pi}{4}} 1 - \cos 2x \, dx \right) \\ = \pi \left[\tan x \right]_0^{\frac{\pi}{4}} - 2\pi \left[\left(x - \frac{1}{2} \sin 2x \right) \right]_0^{\frac{\pi}{4}} \\ = \pi \left[\tan \frac{\pi}{4} - \tan 0 \right] - 2\pi \left[\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right] \\ = \pi [1] - 2\pi \left[\frac{\pi}{4} - \frac{1}{2} \right] \\ = \pi - \frac{2\pi^2}{4} + \pi \\ = 2\pi - \frac{\pi^2}{2}$$

Question 7

$$(a) \quad \alpha + \beta + \gamma = \frac{-b}{a} = \frac{2}{5}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{4}{5}$$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\&= \left(\frac{2}{5}\right)^2 - 2\left(-\frac{4}{5}\right) \\&= \frac{44}{25} \\&= 1 \frac{19}{25}\end{aligned}$$

$$\begin{aligned}(b) (i) \quad \sqrt{3} \cos x - \sin x &= R \cos(x + \alpha) \\&= R(\cos x \cos \alpha - \sin x \sin \alpha)\end{aligned}$$

Equating coeff's $\quad \sqrt{3} = R \cos \alpha$

$$1 = R$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}} \quad \& \quad R = 2$$

$$\therefore \alpha = \frac{\pi}{6} \quad \therefore \sqrt{3} \cos x - \sin x = 2 \cos\left(x + \frac{\pi}{6}\right)$$

$$(ii) \quad 2 \cos\left(x + \frac{\pi}{6}\right) = 1$$

$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

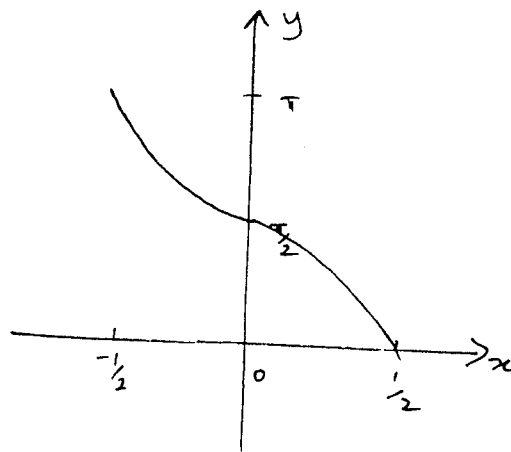
$$x + \frac{\pi}{6} = 2\pi n \pm \frac{\pi}{3}$$

$$x = 2\pi n + \frac{\pi}{6}$$

$$x = \pi$$

Q7 (c)

(i)



$$y = \cos^{-1} x$$

(ii) Domain: $-1 \leq 2x \leq 1$
 $-\frac{1}{2} \leq x \leq \frac{1}{2}$

Range: $0 \leq y \leq \pi$

(iii) $y = \cos^{-1} 2x$

$$y' = \frac{-1}{\sqrt{1-(2x)^2}} \cdot 2$$

$$= \frac{-2}{\sqrt{1-4x^2}}$$

Cuts y axis when $x = 0$

$\therefore$ At $x = 0$, $y = \frac{\pi}{2}$, $y' = \frac{-2}{1} = -2$

$\therefore$ Equation of tangent is

$$y - \frac{\pi}{2} = -2(x - 0)$$

$$y - \frac{\pi}{2} = -2x$$

$$\text{or } y = -2x + \frac{\pi}{2}$$

$$(0) (i) y = \cos^4 x$$

$$\text{let } u = \cos x \quad \therefore y = u^4$$

$$\frac{du}{dx} = -\sin x \quad y = u^4$$

$$\frac{dy}{dx} = 4 \cos^3 x \cdot -\sin x$$

$$= -4 \sin x \cos^3 x$$

$$(ii) \int_0^{\pi/4} \sin x \cos^3 x \, dx$$

$$= -\frac{1}{4} \int_0^{\pi/4} 4 \sin x \cos^3 x \, dx$$

$$= -\frac{1}{4} \left[\cos^4 x \right]_0^{\pi/4}$$

$$= -\frac{1}{4} \left[\left(\frac{1}{\sqrt{2}} \right)^4 - 1 \right]$$

$$= -\frac{1}{4} \left[\frac{1}{2} - 1 \right]$$

$$= -\frac{1}{4} \times -\frac{3}{4}$$

$$= \frac{3}{16}$$